Installing Windows 11 in Hyper-V

Student Name: Mubeen Khan

Lab Environment: CloudLabs VM (Browser-based)

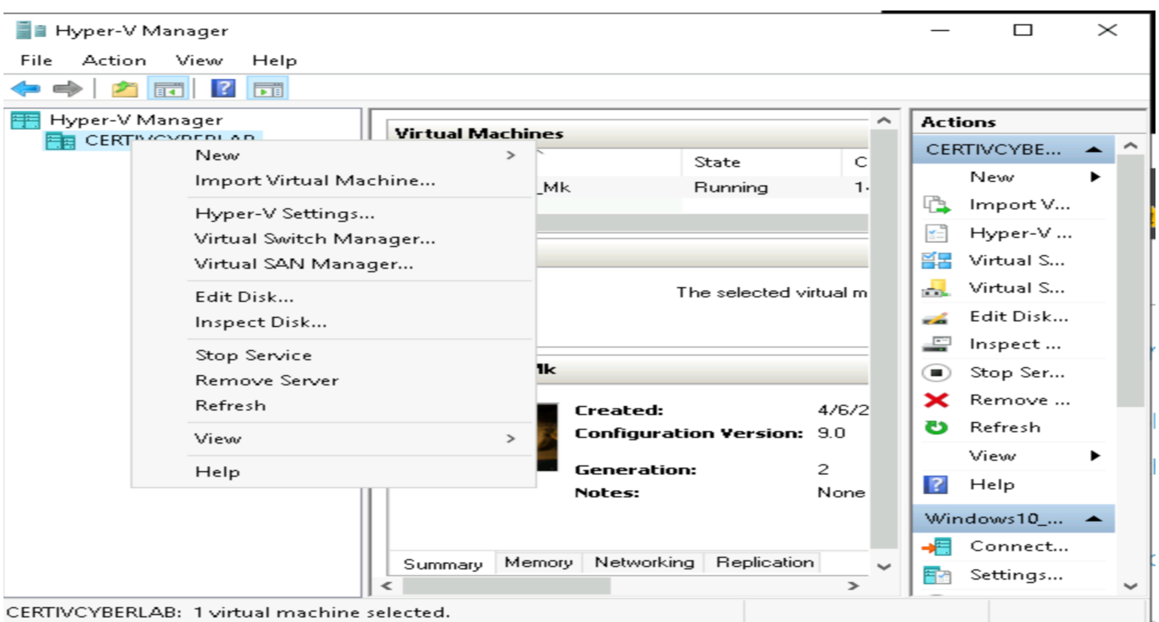
Lab Objective

This lab is part of my Certificate IV in Cyber Security from TAFE NSW. The objective was to setup Windows 11 in Hyper-V.

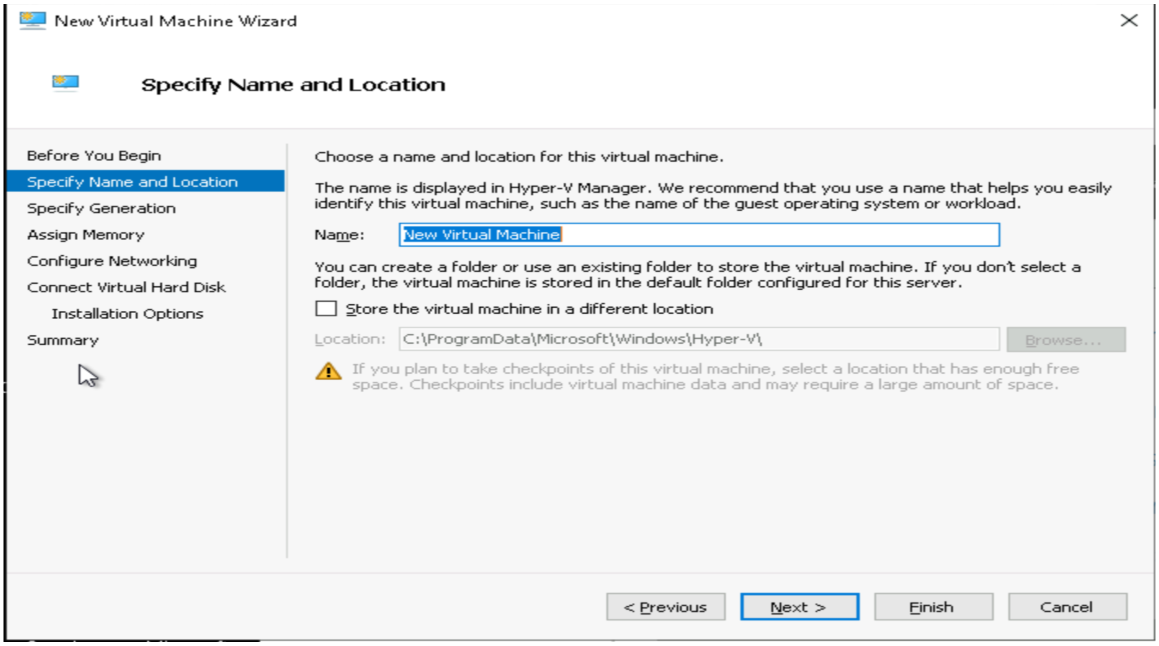
Task 1: Creating a Nested Virtual Machine

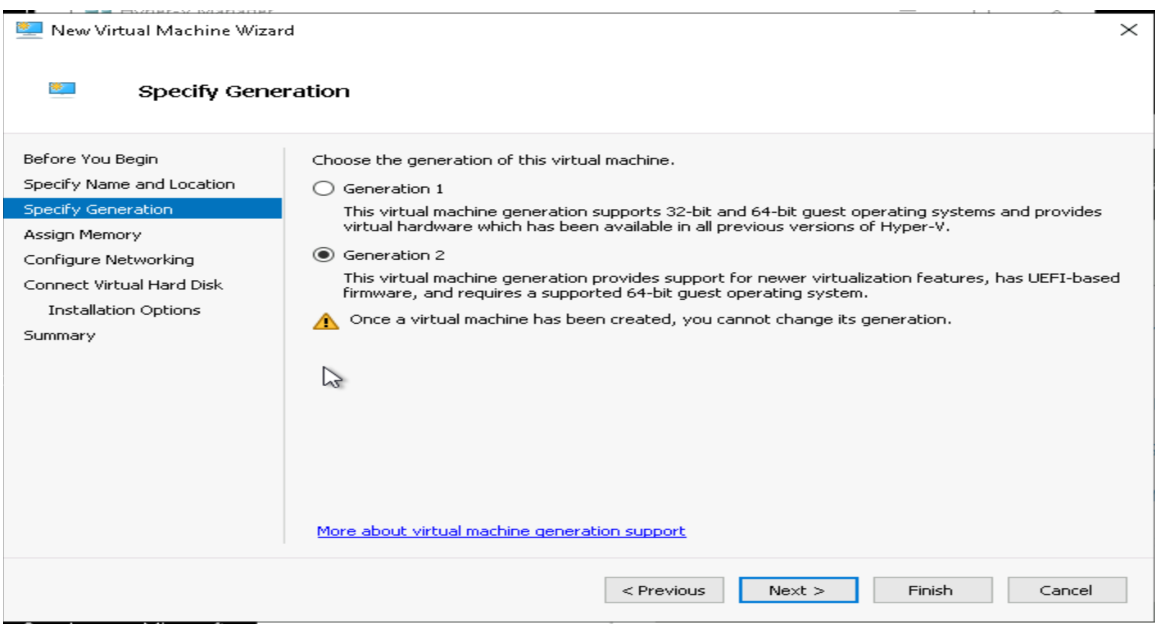
Step 1: Creating a Nested Virtual Machine

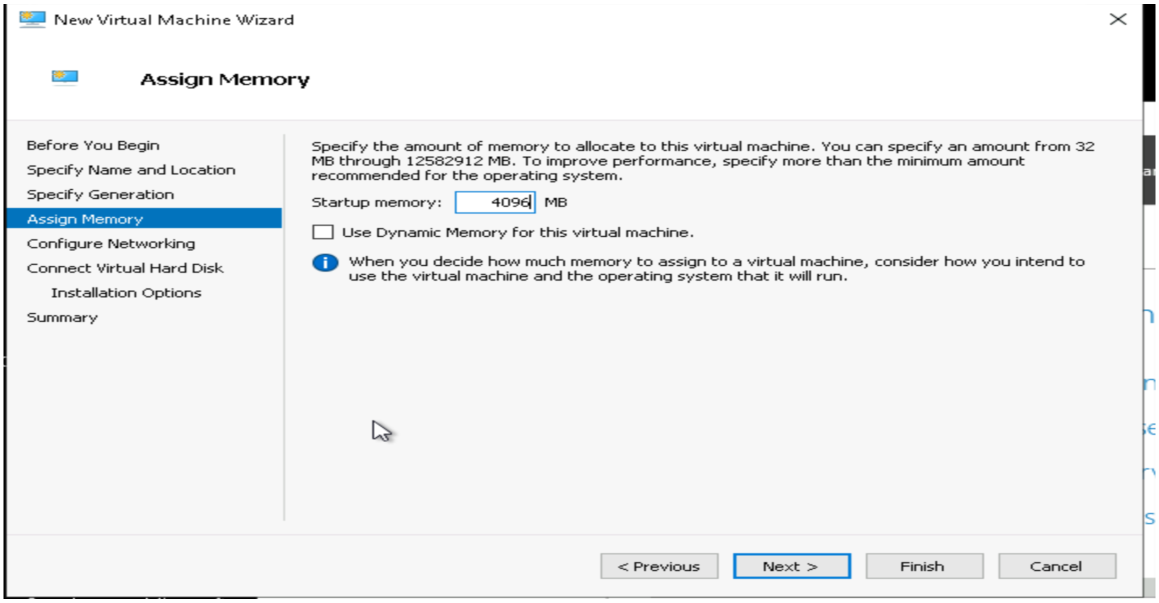
Open Hyper-V Manager → Right click on your server → New → Virtual Machine

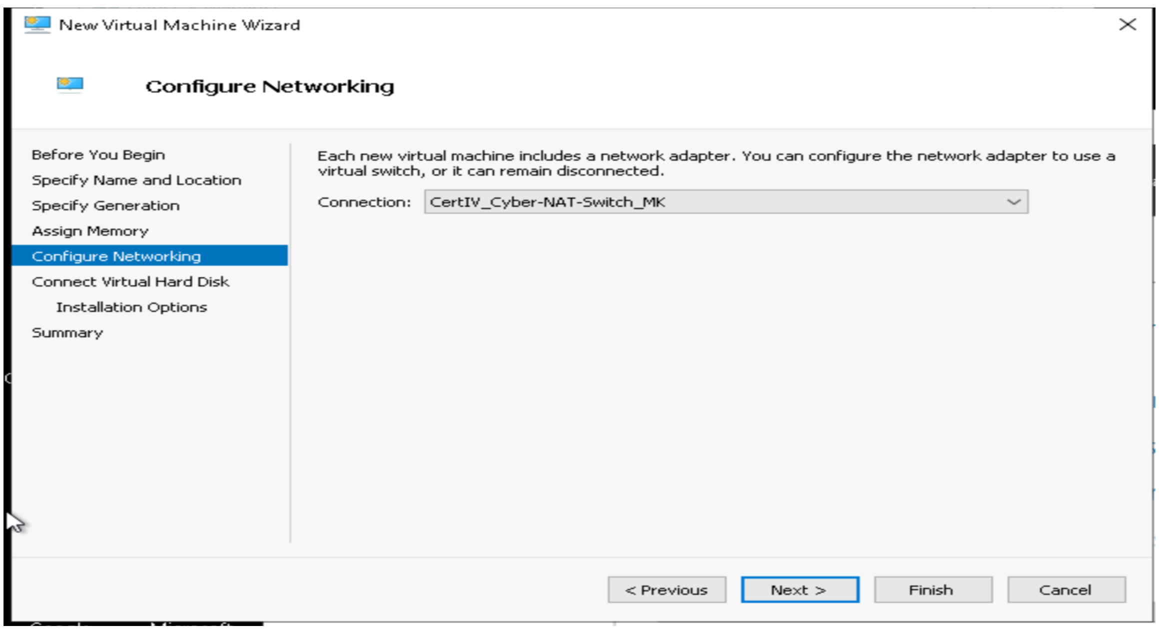


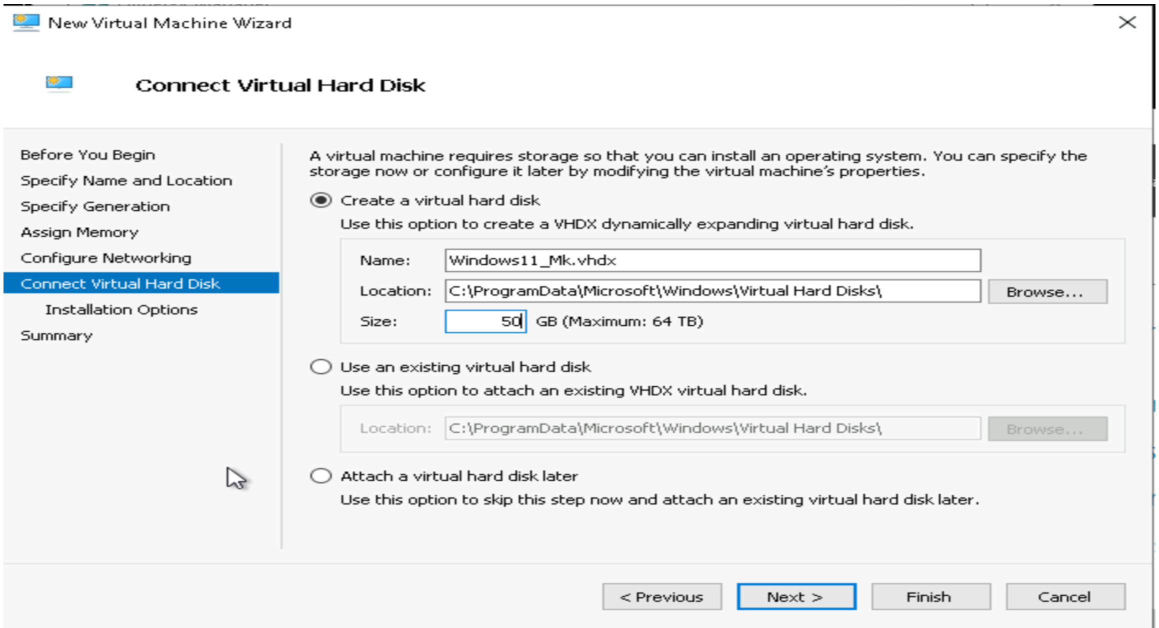
Name your VM → Next → Select generation 2 → Next → Assign startup memory 4096 MB → Next → Select the NAT switch (that I setup in Lab 2) → Next → Create virtual hard disk, 50GB → Next → Select install an operating system form a bootable image file → Click browse → Navigate to Windows 11 ISO file → Open → Finish

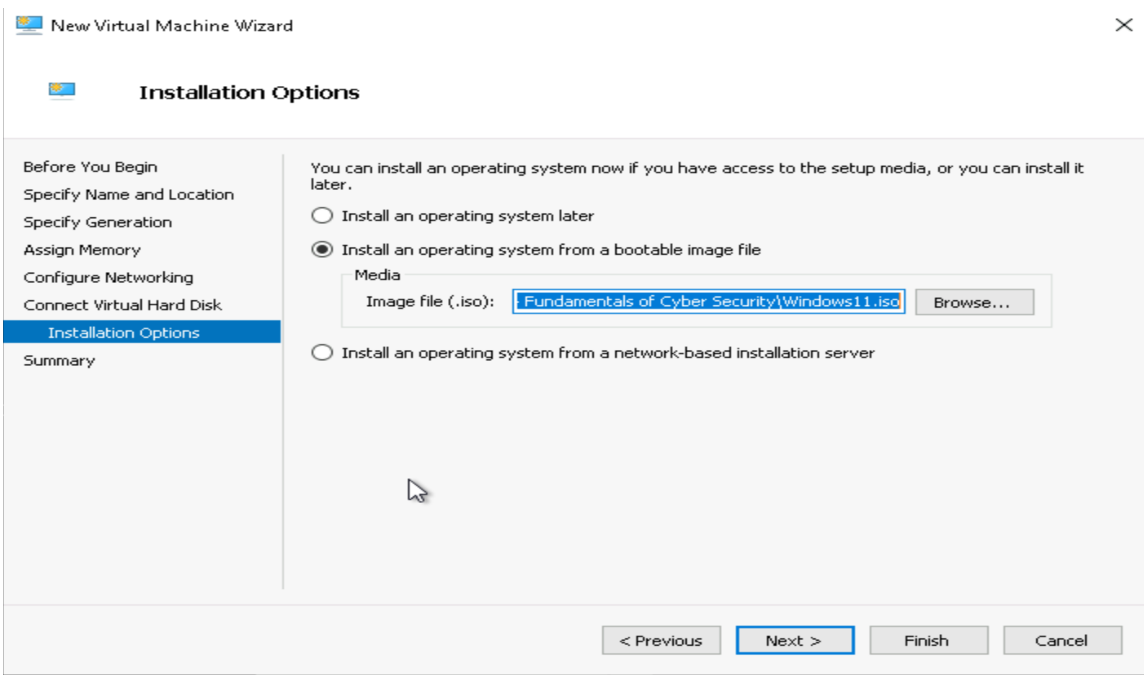

New Virtual Machine Wizard
Specify Name and Location
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Choose a name and location for this virtual machine.
The name is displayed in Hyper-V Manager. We recommend that you use a name that helps you easily identify this virtual machine, such as the name of the guest operating system or workload.
Name: New Virtual Machine
You can create a folder or use an existing folder to store the virtual machine. If you don't select a folder, the virtual machine is stored in the default folder configured for this server.
Store the virtual machine in a different location
Location: C:\ProgramData\Microsoft\Windows\Hyper-V\
Browse...
If you plan to take checkpoints of this virtual machine, select a location that has enough free space. Checkpoints include virtual machine data and may require a large amount of space.
< Previous
Next >
Finish
Cancel


New Virtual Machine Wizard
Specify Generation
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Choose the generation of this virtual machine.
Generation 1
This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.
Generation 2
This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.
Once a virtual machine has been created, you cannot change its generation.
More about virtual machine generation support
< Previous
Next >
Finish
Cancel


New Virtual Machine Wizard
Assign Memory
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 12582912 MB. To improve performance, specify more than the minimum amount recommended for the operating system.
Startup memory: 4096 MB
Use Dynamic Memory for this virtual machine.
When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.
< Previous
Next >
Finish
Cancel


New Virtual Machine Wizard
Configure Networking
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Each new virtual machine includes a network adapter. You can configure the network adapter to use a virtual switch, or it can remain disconnected.
Connection: CertIV_Cyber-NAT-Switch_MK
< Previous
Next >
Finish
Cancel

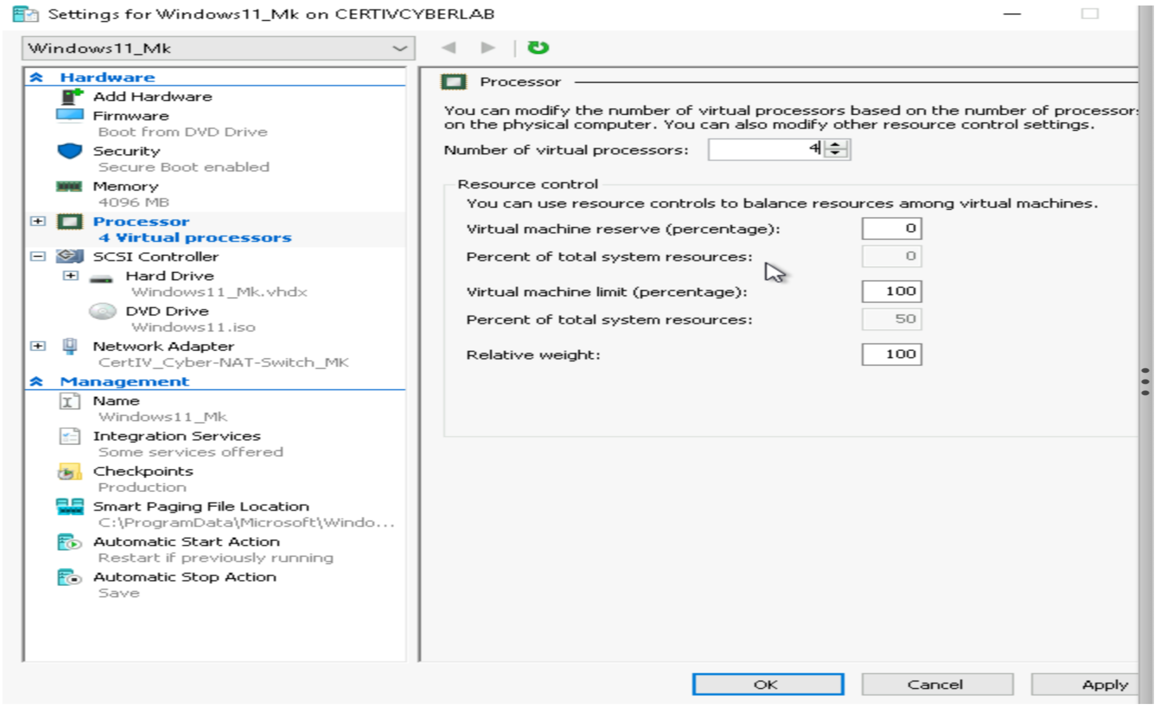


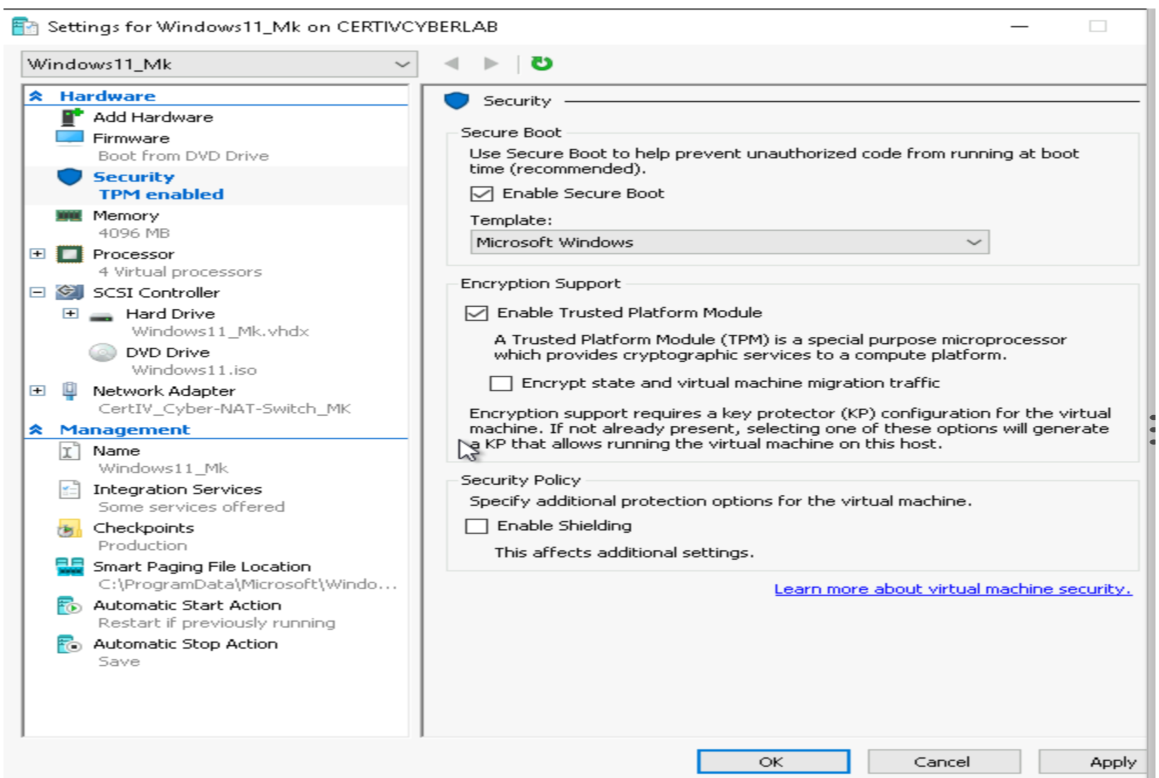


Step 2: Configuring the Nested Virtual Machine

Open the VM → Right click on the new VM → Settings → Increase processor to 4 and click apply → Enable TPM by clicking on security → Enable Trusted Platform Module → Apply

Enable Guest Services → click integration service → check guest services → Apply → Ok


Settings for Windows11_Mk on CERTIVCYBERLAB
Windows11_Mk
Hardware
Add Hardware
Firmware
Boot from DVD Drive
Security
Secure Boot enabled
Memory
4096 MB
Processor
4 Virtual processors
SCSI Controller
Hard Drive
Windows11_Mk.vhdx
DVD Drive
Windows11.iso
Network Adapter
CertIV_Cyber-NAT-Switch_MK
Management
Name
Windows11_Mk
Integration Services
Some services offered
Checkpoints
Production
Smart Paging File Location
C:\ProgramData\Microsoft\Windo...
Automatic Start Action
Restart if previously running
Automatic Stop Action
Save
Processor
You can modify the number of virtual processors based on the number of processors on the physical computer. You can also modify other resource control settings.
Number of virtual processors: 4
Resource control
You can use resource controls to balance resources among virtual machines.
Virtual machine reserve (percentage): 0
Percent of total system resources: 0
Virtual machine limit (percentage): 100
Percent of total system resources: 50
Relative weight: 100
OK
Cancel
Apply


Settings for Windows11_Mk on CERTIVCYBERLAB
Windows11_Mk
Hardware
Add Hardware
Firmware
Boot from DVD Drive
Security
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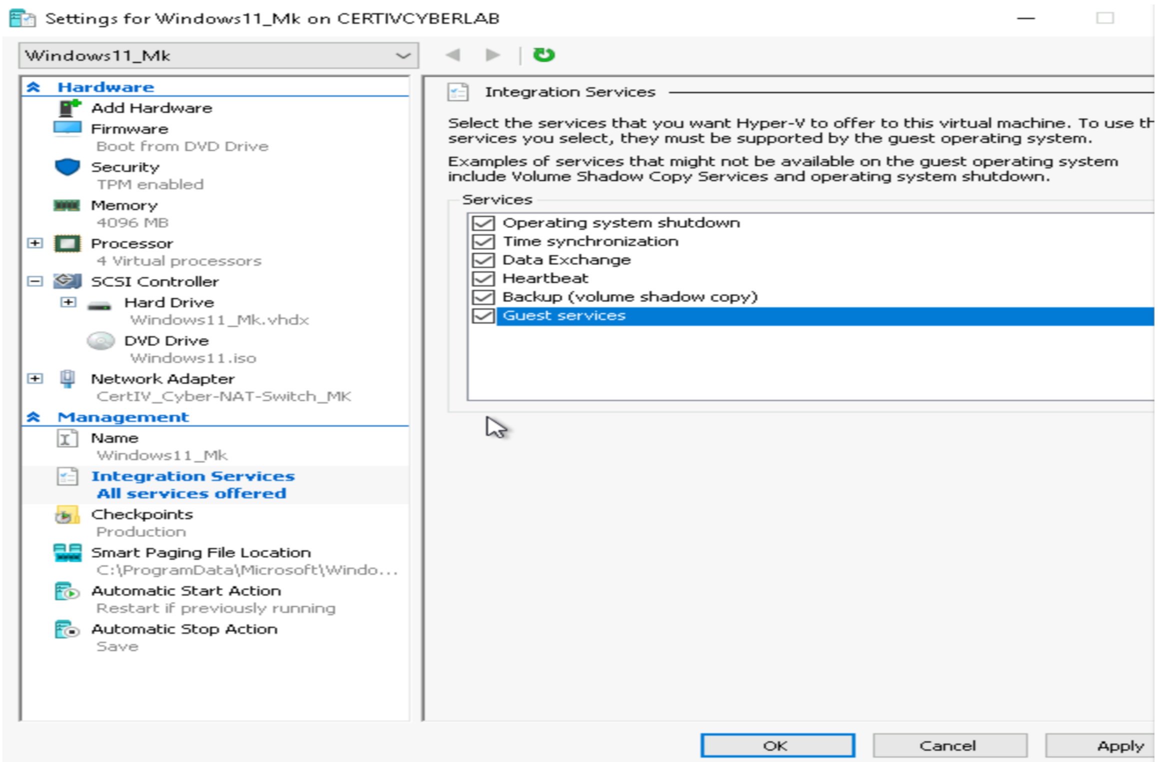
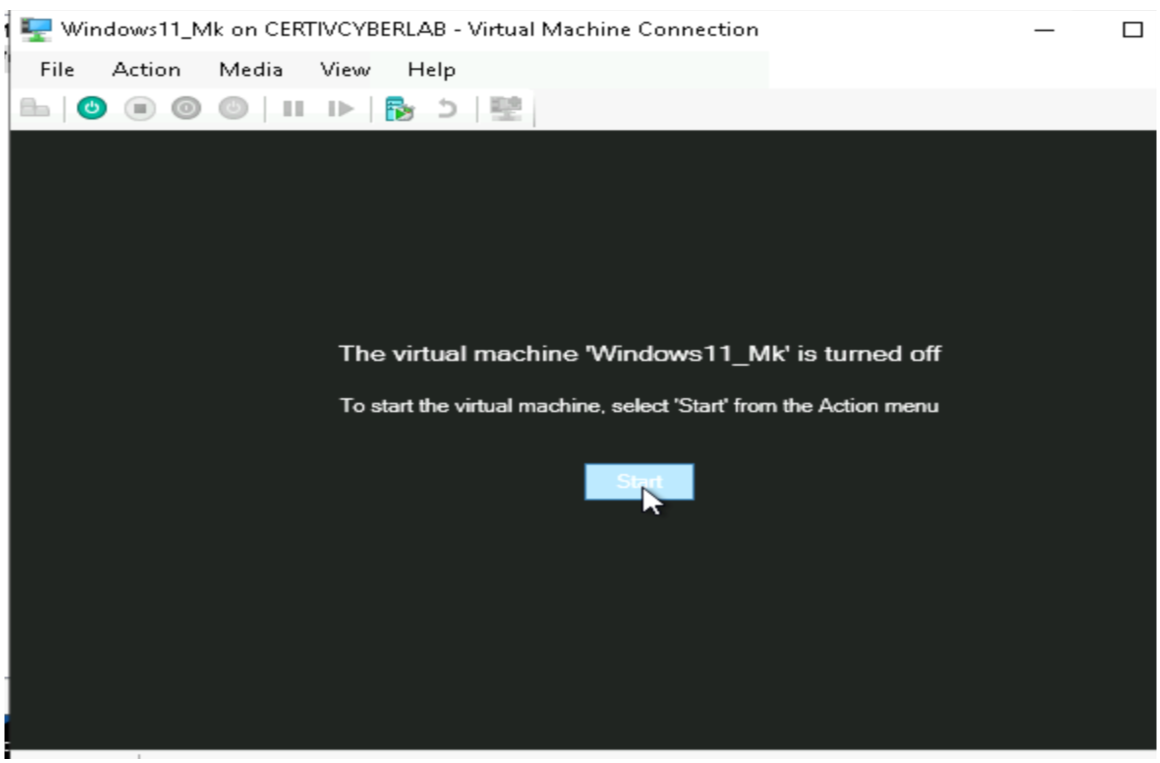
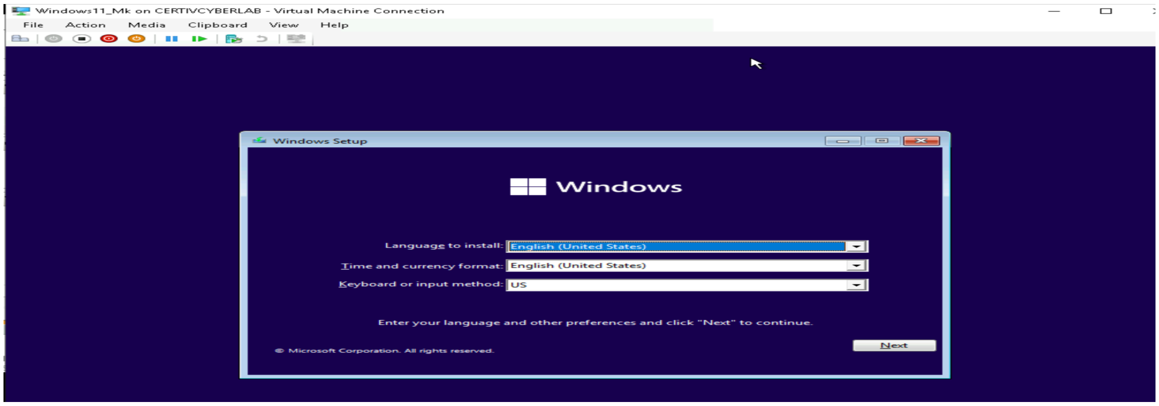
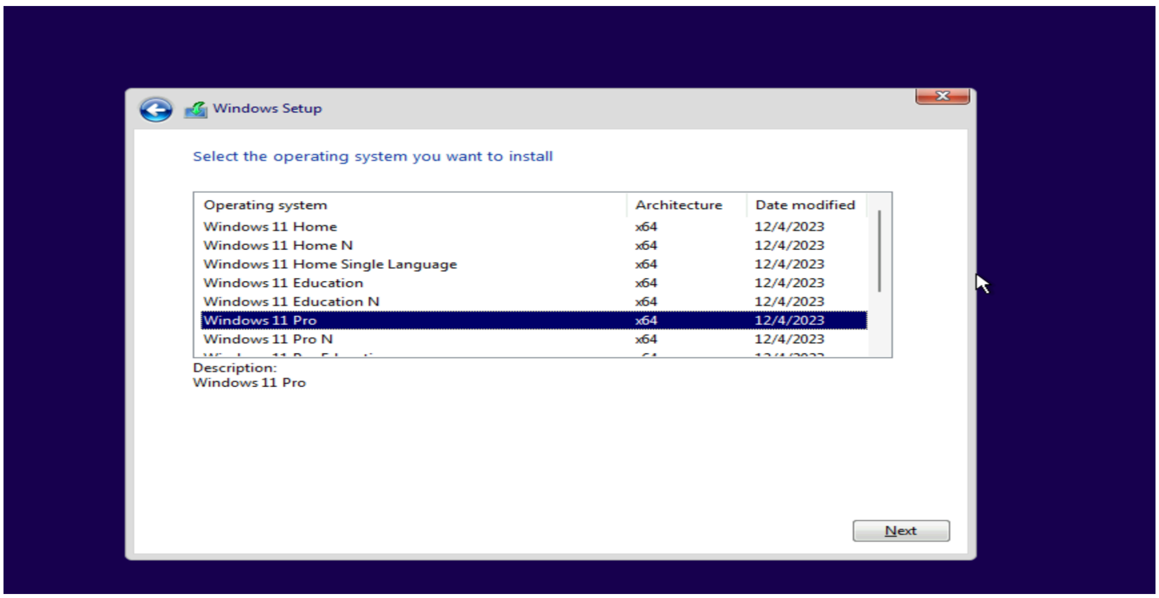
4096 MB
Processor
4 Virtual processors
SCSI Controller
Hard Drive
Windows11_Mk.vhdx
DVD Drive
Windows11.iso
Network Adapter
CertIV_Cyber-NAT-Switch_MK
Management
Name
Windows11_Mk
Integration Services
Some services offered
Checkpoints
Production
Smart Paging File Location
C:\ProgramData\Microsoft\Windo...
Automatic Start Action
Restart if previously running
Automatic Stop Action
Save
Security
Secure Boot
Use Secure Boot to help prevent unauthorized code from running at boot time (recommended).
Enable Secure Boot
Template:
Microsoft Windows
Encryption Support
Enable Trusted Platform Module
A Trusted Platform Module (TPM) is a special purpose microprocessor which provides cryptographic services to a compute platform.
Encrypt state and virtual machine migration traffic
Encryption support requires a key protector (KP) configuration for the virtual machine. If not already present, selecting one of these options will generate a KP that allows running the virtual machine on this host.
Security Policy
Specify additional protection options for the virtual machine.
Enable Shielding
This affects additional settings.
Learn more about virtual machine security.
OK
Cancel
Apply

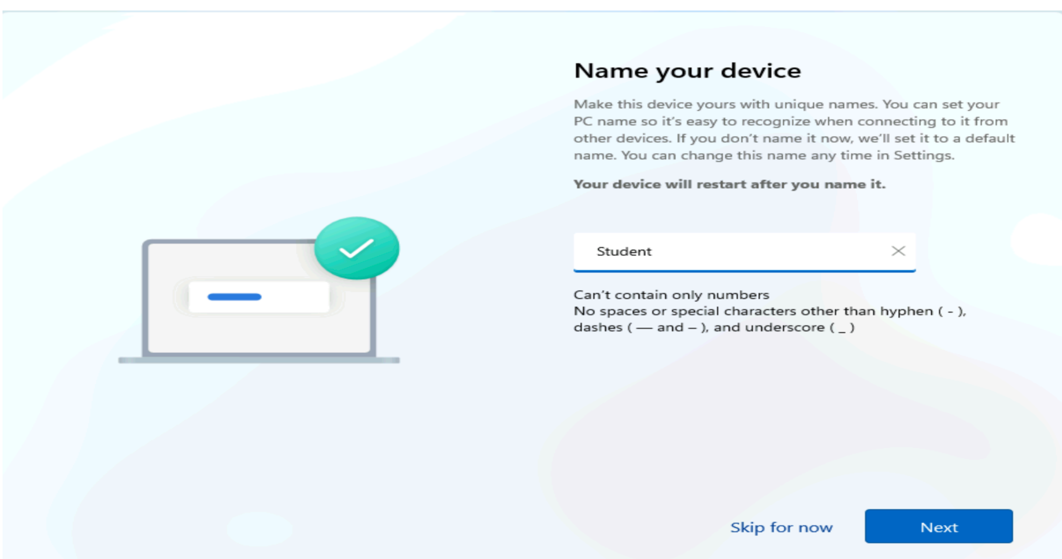


Step 3: Opening and setting up the Windows 11 VM

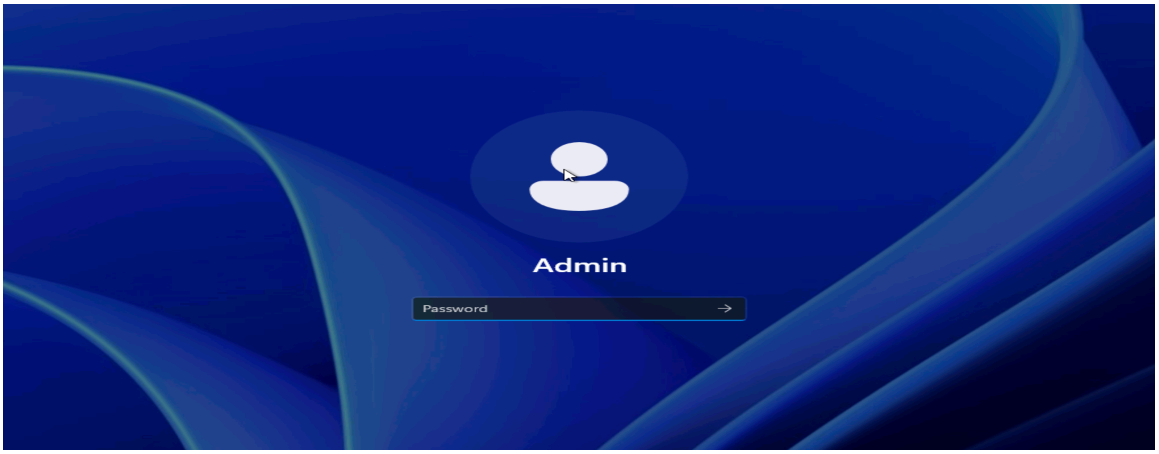


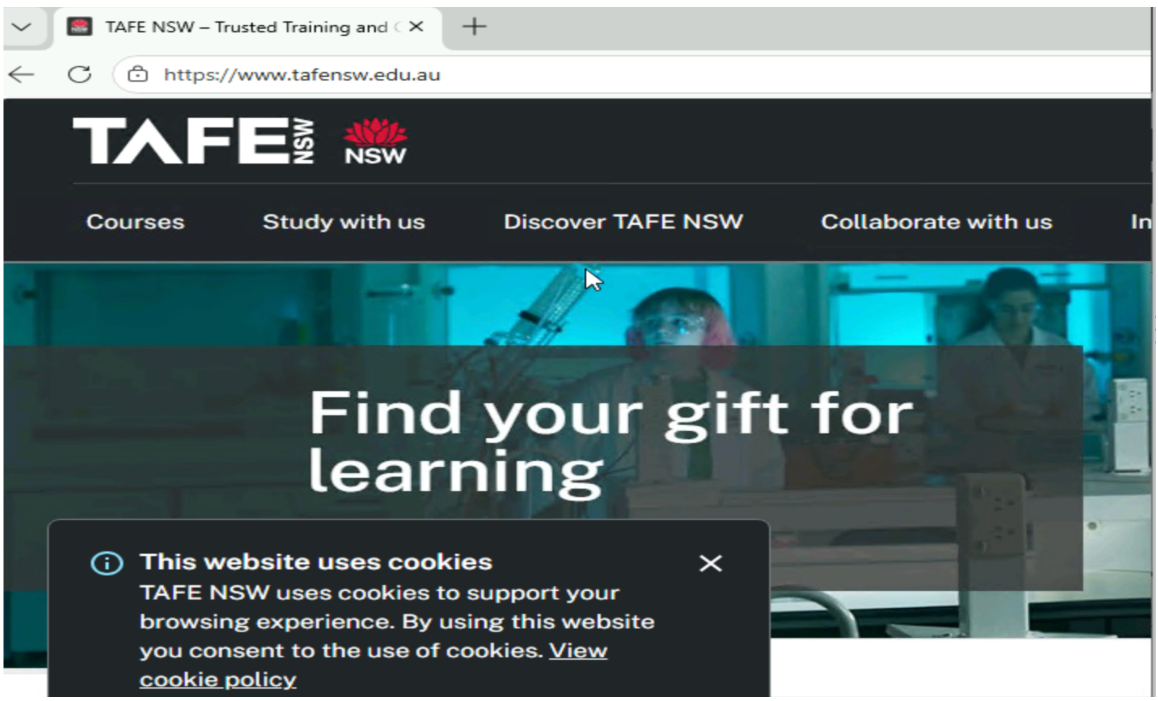






Step 4: Testing Internet Connectivity


Admin
Password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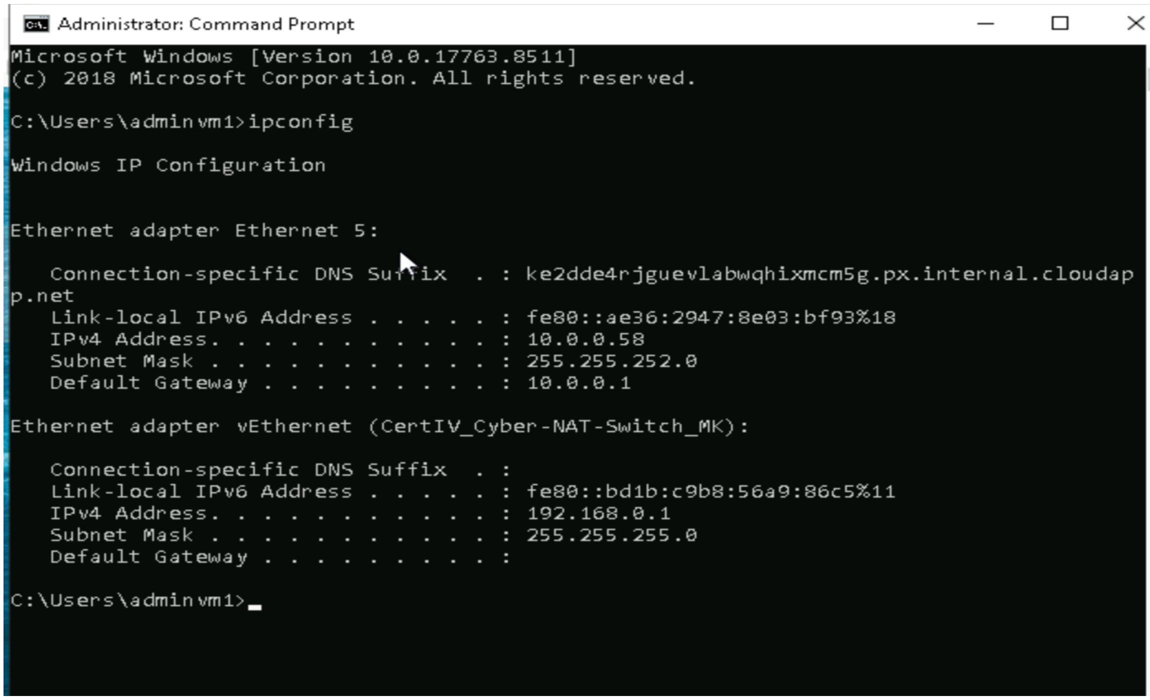
Administrator: Command Prompt
Microsoft Windows [Version 10.0.17763.8511]
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C:\Users\adminvm1>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet 5:
Connection-specific DNS Suffix . : ke2dde4rjguevlabwqhixmcm5g.px.internal.cloudapp.net
Link-local IPv6 Address . . . . . : fe80::ae36:2947:8e03:bf93%18
IPv4 Address. . . . . : 10.0.0.58
Subnet Mask . . . . : 255.255.252.0
Default Gateway . . . . : 10.0.0.1
Ethernet adapter vEthernet (CertIV_Cyber-NAT-Switch_MK):
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::bd1b:c9b8:56a9:86c5%11
IPv4 Address. . . . . : 192.168.0.1
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
C:\Users\adminvm1>